

From Code to the Real World

ESP32 • IoT • APIs • AI • Open Hardware

CHITCHAT / DEMO DAY

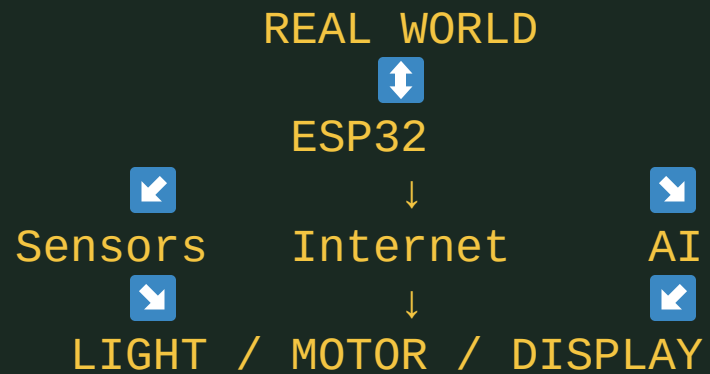
Zero experience required

What if your code could...

- Know when a room is too hot?
- Detect someone entering?
- Know that rain is coming?
- React when your football team scores?
- Ask AI what to do next?
- Turn on a light, motor, fan or pump?

Software doesn't have to stay inside a screen.

The Big Idea



Sense → Think → Connect → Act

What is Open Hardware?

Hardware we can:

- Learn from
- Program
- Connect
- Modify
- Extend
- Build products around

Examples:

Arduino • ESP8266 • ESP32 • Raspberry Pi • M5Stack • XIAO • LilyGO

Meet ESP32

A tiny programmable computer with connectivity built in.

ESP32

- CPU + Memory
- Wi-Fi
- Bluetooth
- GPIO
- ADC
- PWM
- I²C
- SPI
- UART

One board can connect software to the physical world.

The ESP Family

Arduino ecosystem

ESP8266

ESP32

ESP32

ESP32-C3

ESP32-S3

ESP32-C6

And ready-to-build devices:

M5Stack • Seeed XIAO • LilyGO • many more

You Can Start Without Hardware

Playground → Simulator → Real Device

MIT Scratch / Blocks



Programming Logic



Wokwi



Virtual ESP32



Real ESP32

Start in the browser. Move to hardware when ready.

From Blocks to Professional Development

Scratch / Block Programming



Wokwi



Arduino IDE



MicroPython



VS Code + PlatformIO



ESP-IDF

You don't need to learn everything on day one.

Demo 1 --- The Simplest Device

```
graph TD; BUTTON --> ESP32; ESP32 --> LED;
```

BUTTON
↓
ESP32
↓
LED

No AI.

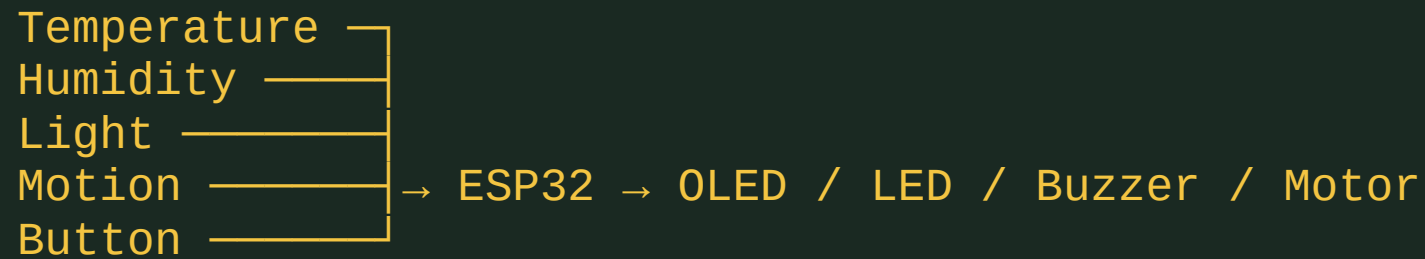
No cloud.

No Internet.

INPUT → PROCESS → OUTPUT

This is the foundation of almost everything we build.

Demo 2 --- Let It Sense the World



Examples:

- Hot room → fan
- Dark room → light
- Motion → alert
- Dry soil → pump

Add Wi-Fi

```
Sensor
  ↓
ESP32
  ↓ Wi-Fi
Phone Browser
```

Now your phone can:

- Read sensor values
- Control an LED
- Start a motor
- Change settings

Welcome to IoT.

Local Data + Global Data

Local --- around us Global --- Internet

Temperature Weather
Humidity Football
Motion Currency
Light Air quality
Soil moisture Transport
Distance News
Buttons Maps

ESP32 can use both.

Demo 3 --- Weather Becomes Physical

Weather API



Internet



ESP32



Melbourne

17°C

Rain: 85% 



BLUE LED + "Take Umbrella"

Global information → physical action.

A More Fun API Demo

Football API



GOAL!



ESP32



LED Buzzer Servo



Team Flag

Something happens somewhere in the world...

...and hardware on our desk reacts.

What Is Happening Behind the Demo?

```
ESP32
  ↓
Wi-Fi
  ↓
HTTP Request
  ↓
API
  ↓
JSON Data
  ↓
ESP32
  ↓
ACTION
```

These are not just web-development concepts.

They can control the real world.

Can ESP32 Use AI?

Yes.

```
graph TD; ESP32 --> Internet; Internet --> AIService[AI API / AI Service]; AIService --> Answer[Answer / Decision]; Answer --> ESP32_2[ESP32]; ESP32_2 --> Action[Physical Action];
```

ESP32 handles the device.

AI can help understand, classify or decide.

Demo 4 --- Sensor → AI → Action

Temperature: 31°C

Humidity: 82%

↓

AI

↓

"Hot and very humid.
Turn ventilation on."

↓

ESP32 → FAN

Local sensor data can become context for AI.

And the Other Direction...

YOU



"Turn the light on"



AI / Voice Service



ESP32



💡 LIGHT

Or:

"Set brightness to 50%"



ESP32



PWM

AI can trigger physical actions.

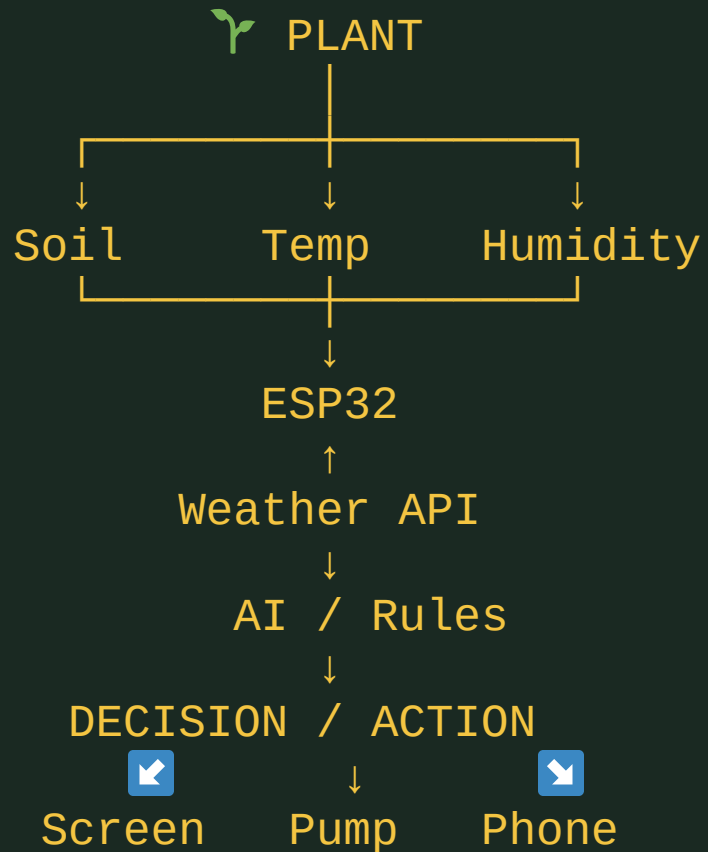
Give the Device Eyes

```
graph TD; A[ESP32 Camera] --> B[Image]; B --> C[Vision AI]; C --> D["Plant leaf may be unhealthy"]; D --> E[ESP32]; E --> F[Display / Alert / Action]
```

Camera + network + AI turns a small board into a smart sensor.

The Big WOW Demo

Smart Plant / Smart Environment



Simple Automation vs Smart Decision

Local data:

```
Soil = DRY  
Temperature = 29°C  
Humidity = 48%
```

Global data:

```
Weather forecast:  
Rain expected in 2 hours
```

Simple automation:

```
IF soil == dry → WATER
```

Smarter decision:

```
"Don't water yet. Rain is expected soon."
```

From Breadboard to Product

We start here:

ESP32 + Breadboard + Wires + Sensors

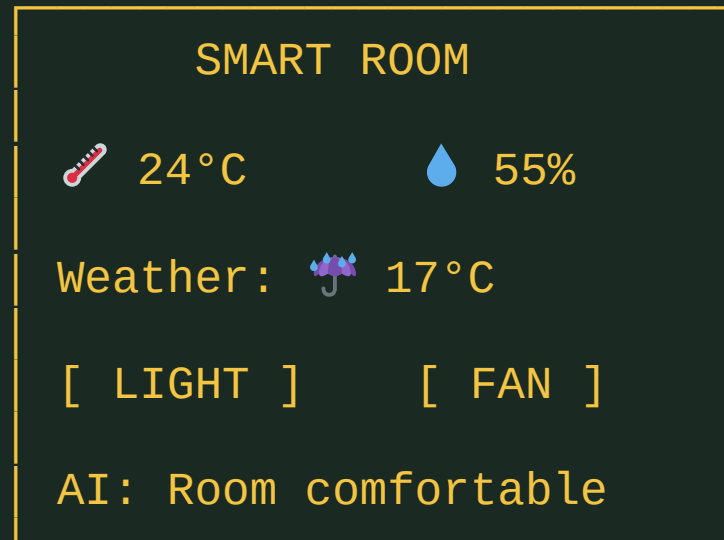
And can progress to:

M5Stack / Integrated ESP32 Device

- Touchscreen
- Wi-Fi
- Sensors
- Speaker / Microphone
- Camera
- Battery

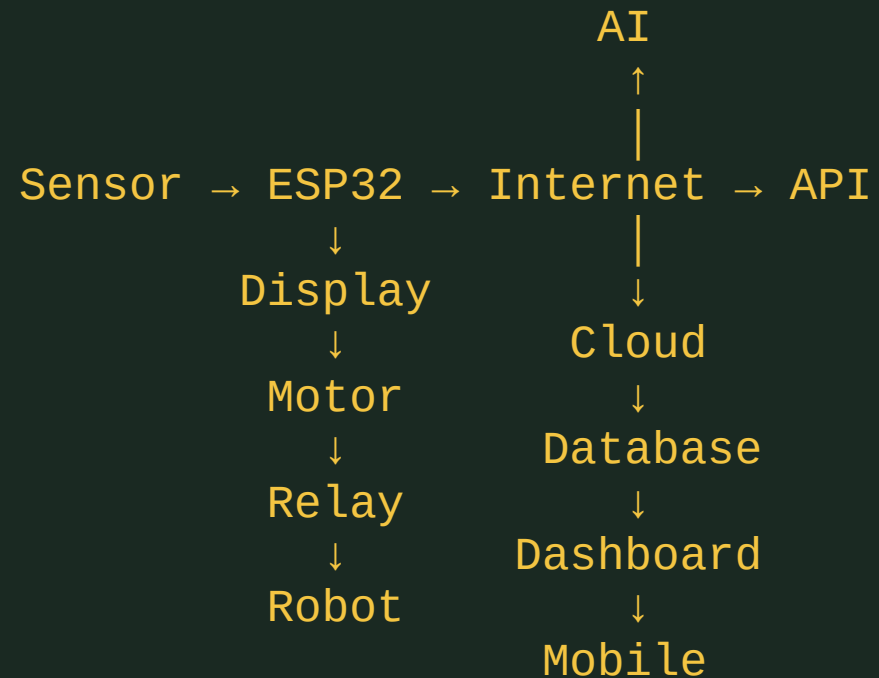
Same concepts --- increasingly polished hardware.

Example: Smart Room Controller



A student project can start looking like a real product.

The Complete World



Hardware + Software + Internet + AI.

What Will the Bootcamp Teach?

ZERO → LOGIC → ELECTRONICS



ESP8266 / ESP32 → SENSORS



INTERNET → APIs → AI



AUTOMATION → PRODUCT

What Will You Actually Touch?

A first kit

ESP32 · breadboard · LEDs · buttons · resistors · sensors

Start with the physical basics.

Build → break → debug → understand → rebuild.

Then Grow the Project

- OLED display
- Buzzer
- Servo
- Relay
- Camera
- M5Stack

From a first LED to a product-like device.

Where Can This Lead?

- Smart home
- Smart agriculture
- Environmental monitoring
- Robotics
- Wearables
- Industrial IoT
- Home Assistant
- TinyML / Edge AI
- AI-connected devices
- Custom PCB / product development

Your first LED can be the first step toward a real product.

Today: Watch Us Build It

Bootcamp: Build It Yourself

```
graph TD; CODE --> LED; LED --> SENSOR; SENSOR --> REAL_WORLD[REAL WORLD]; REAL_WORLD --> INTERNET; INTERNET --> GLOBAL_DATA[GLOBAL DATA]; GLOBAL_DATA --> AI; AI --> PHYSICAL_ACTION[PHYSICAL ACTION]; PHYSICAL_ACTION --> REAL_PRODUCT[REAL PRODUCT];
```

CODE
↓
LED
↓
SENSOR
↓
REAL WORLD
↓
INTERNET
↓
GLOBAL DATA
↓
AI
↓
PHYSICAL ACTION
↓
REAL PRODUCT

Questions?

WHAT WOULD YOU BUILD IF YOUR CODE COULD CONTROL THE REAL WORLD?

VibeCode.Tours × Open Hardware

Demo Day: Build Together

Three projects. One clear journey.

SCRATCH LOGIC → WOKWI SIMULATION → API EVENT BOX

One project is hands-on. Two projects are visual WOW demos.

Guest Project --- Touch Light

We ask the room:

"What should happen when I touch this sensor?"

Touch Sensor



ESP8266



LED

Guests help decide the behaviour before we write any code.

No hardware? Run the same logic as a Wokwi simulation.

Think Together → Write Together

First, write the idea in plain language:

```
IF touched
    light ON
ELSE
    light OFF
```

Then turn the idea into code together.

```
IDEA
↓
LOGIC
↓
CODE
↓
UPLOAD
↓
REAL-WORLD RESULT 💡
```

Make It Interactive

Once the first version works, ask the audience:

"What should we change?"

Possible ideas:

- Touch once → ON
- Touch again → OFF
- Blink three times
- Add a sensor condition
- Trigger something over Wi-Fi

The guests become part of the development process.

Project 2 --- Weather Event Box

PHONE BUTTON / WEATHER API



VIRTUAL ESP32



BLUE LIGHT BUZZER UMBRELLA FLAG

Press **Rain expected** on a phone. The event box reacts on the projected simulator.

Global data → local physical action.

Project 3 --- Football Event Box

Same tiny board. A different event.

PHONE BUTTON / FOOTBALL API



GOAL EVENT



VIRTUAL ESP32



FLASH



SOUND



FLAG

Press Football goal. A goal somewhere else becomes a celebration here.

One Box --- Many Stories

One reusable demo device:



Change the event + logic...

and it becomes a completely different project.

Optional Extension --- Smart Plant 🌱

SOIL IS DRY + RAIN IS COMING



SMART DECISION



WAIT / WATER / WARN

Mention this as the next bootcamp project, not a fourth main Demo Day build.

One Box --- Many Ideas

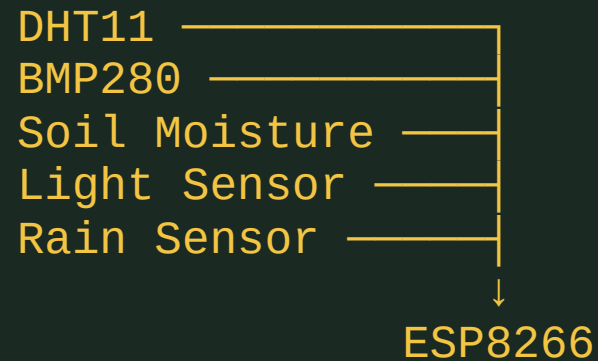
Weather	→	
Football	→	
Air Quality	→	
Transport	→	
GitHub Event	→	
Smart Plant	→	
AI Decision	→	

The hardware can stay almost the same.

The idea + data + code changes the product.

Add Our Local Sensors

Use the modules we already have:



Now the device understands what is happening **here**.

Local + Global

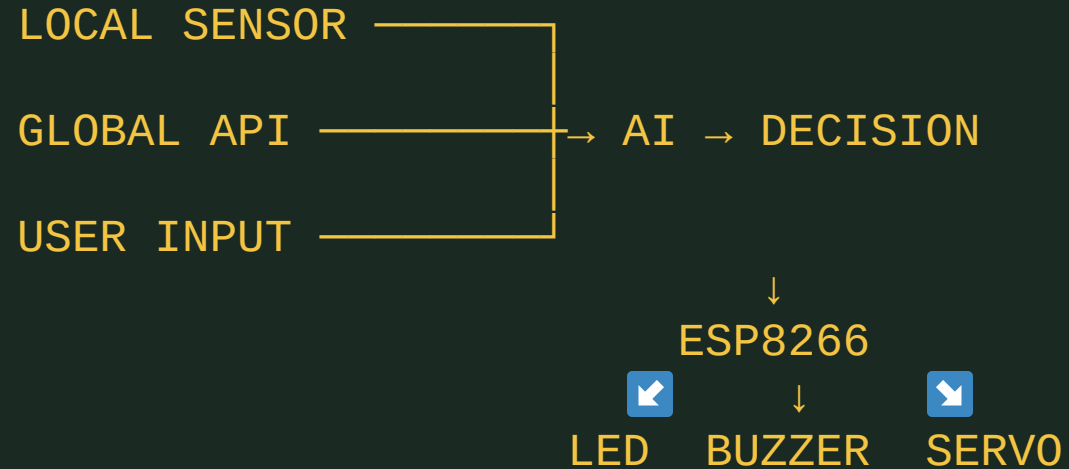


Example:

Plant is dry locally + rain is coming globally.

What should we do?

Now Add AI



AI becomes another tool in the project --- not the whole project.

WOW Example --- Smart Decision

LOCAL + GLOBAL DATA



AI



"Take a jacket and umbrella."



ESP8266 → LED + BUZZER + SERVO

What Did We Just Learn?

INPUT / API → ESP8266 → OUTPUT

Same foundation. More inputs and connectivity create bigger projects.

Read → decide → act.

Demo Day Journey

WATCH



THINK TOGETHER



WRITE TOGETHER



BUILD TOGETHER



SEE IT WORK



TOUCH LED



WEATHER + FOOTBALL WOW



"WHAT CAN I BUILD?"

That question leads directly into the Bootcamp.

Today We Build One Together

In the Bootcamp, you build your own.

ESP8266 / ESP32 • Sensors • APIs • IoT • AI • Real Projects